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(54) **SELECTIVE USE OF DIFFERENT ADVANCED INTERFACE BUS WITH ELECTRONIC CHIPS**

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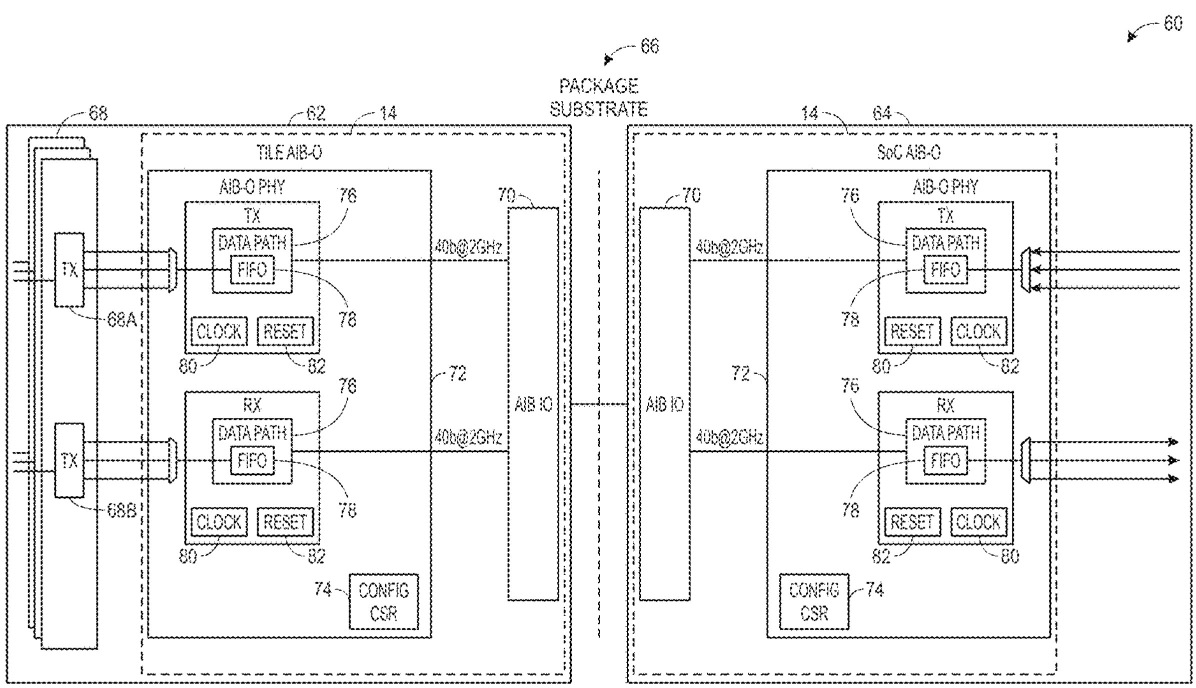
(57) **ABSTRACT**

A digitally communicative circuit may use standardized interfaces for connection and communication with other circuit components. Such digitally communicative circuit may benefit from using wider variety of interconnect schemes with the respective interfaces for transmission and reception of data. Some chiplets may communicate using a high data bandwidth interface while other chiplets may communicate using interfaces with lower data bandwidth. Alternate interface is introduced that may facilitate scaled communication with Advanced Interface Bus 2.0 without translation circuitry and with different data bandwidth.



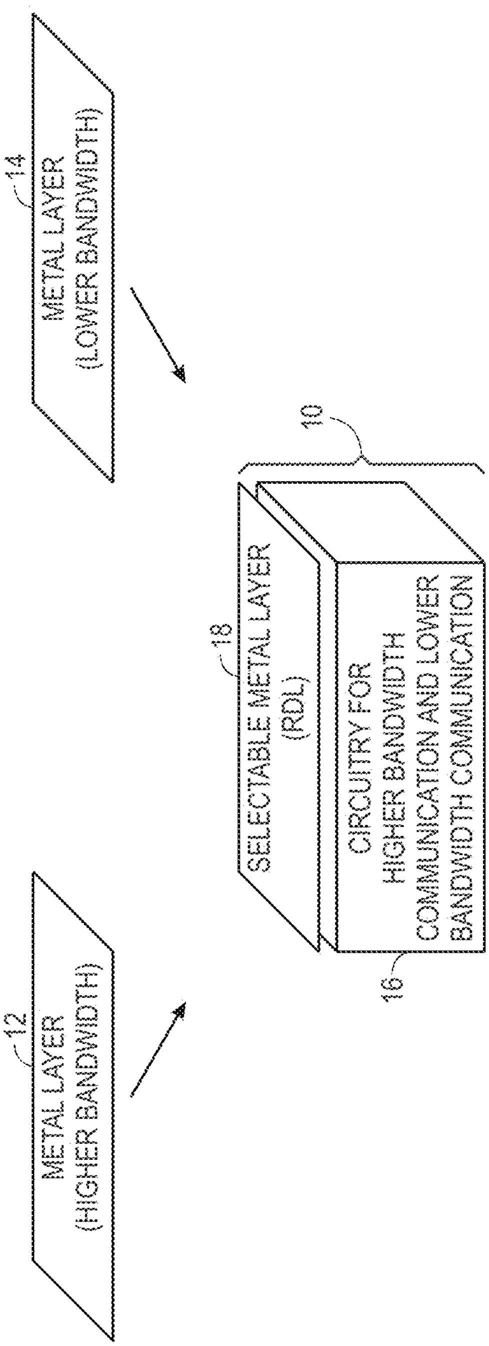
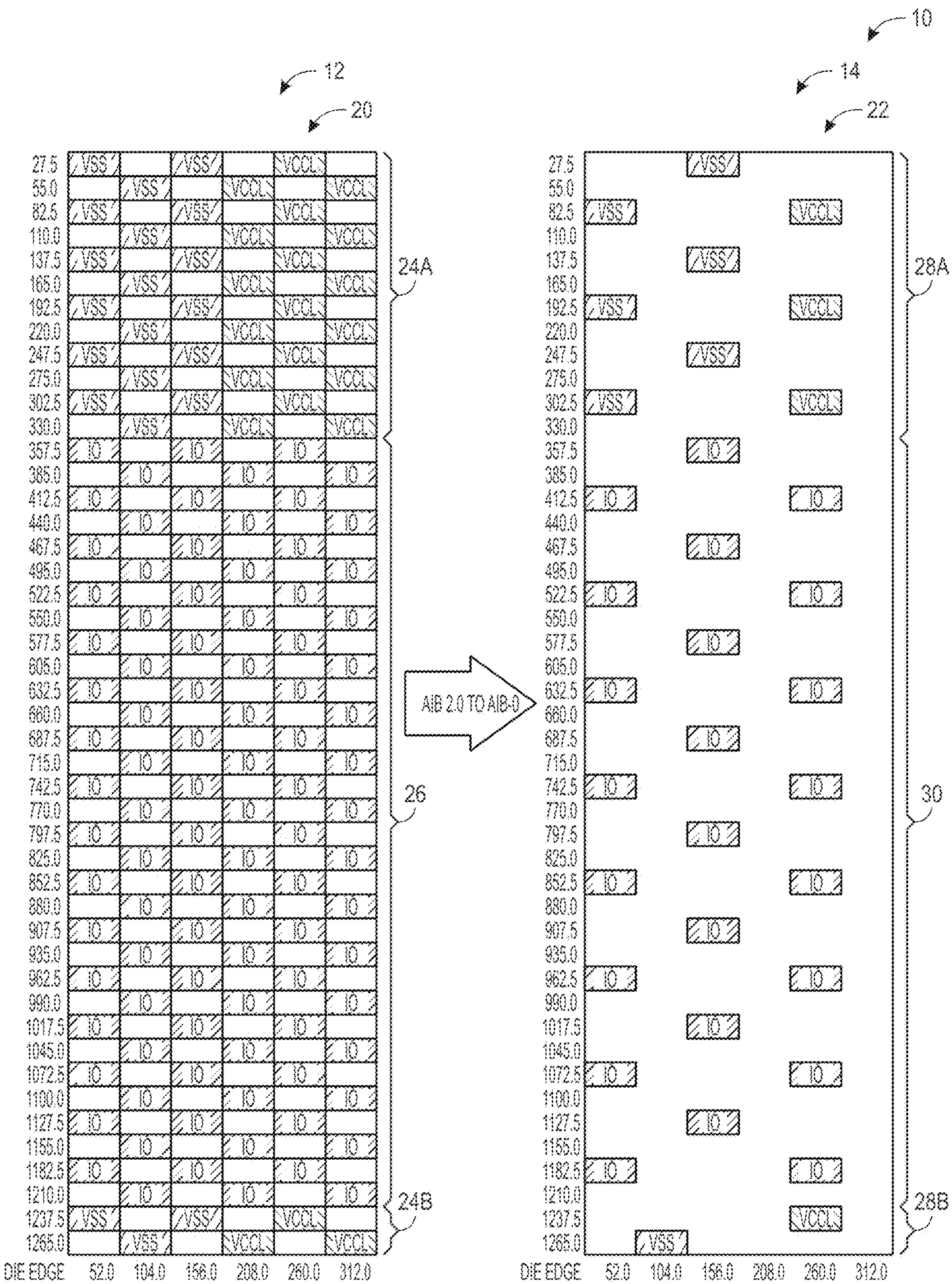

METAL LAYER
(LOWER BANDWIDTH)
14
10
18
SELECTABLE METAL LAYER
(RDL)
CIRCUITRY FOR
HIGHER BANDWIDTH
COMMUNICATION AND LOWER
BANDWIDTH COMMUNICATION
16
12
METAL LAYER
(HIGHER BANDWIDTH)

*FIG. 1*


10
12
20
14
22
24A
26
24B
28A
30
28B
AIB 2.0 TO AIB-0
VSS
VCCL
IO
DIE EDGE
52.0
104.0
156.0
208.0
260.0
312.0
27.5
55.0
82.5
110.0
137.5
165.0
192.5
220.0
247.5
275.0
302.5
330.0
357.5
385.0
412.5
440.0
467.5
495.0
522.5
550.0
577.5
605.0
632.5
660.0
687.5
715.0
742.5
770.0
797.5
825.0
852.5
880.0
907.5
935.0
962.5
990.0
1017.5
1045.0
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1210.0
1237.5
1265.0

FIG. 2

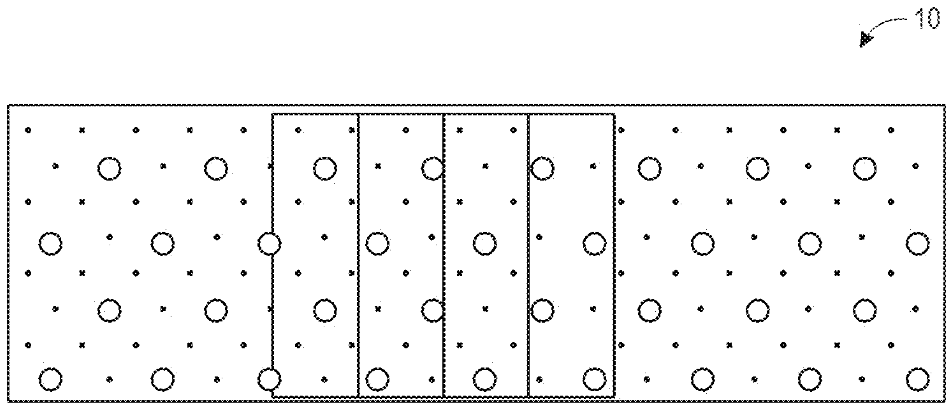
10

*FIG. 3*

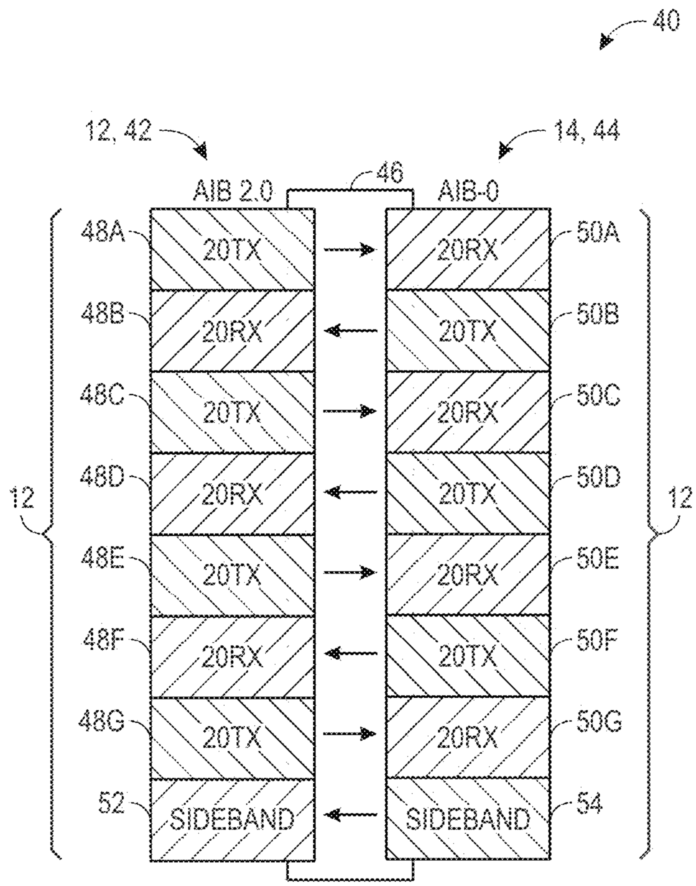
40
12, 42
14, 44
46
AIB 2.0
AIB-0
48A
20TX
20RX
50A
48B
20RX
20TX
50B
48C
20TX
20RX
50C
48D
20RX
20TX
50D
12
12
48E
20TX
20RX
50E
48F
20RX
20TX
50F
48G
20TX
20RX
50G
52
SIDEBAND
SIDEBAND
54

*FIG. 4*

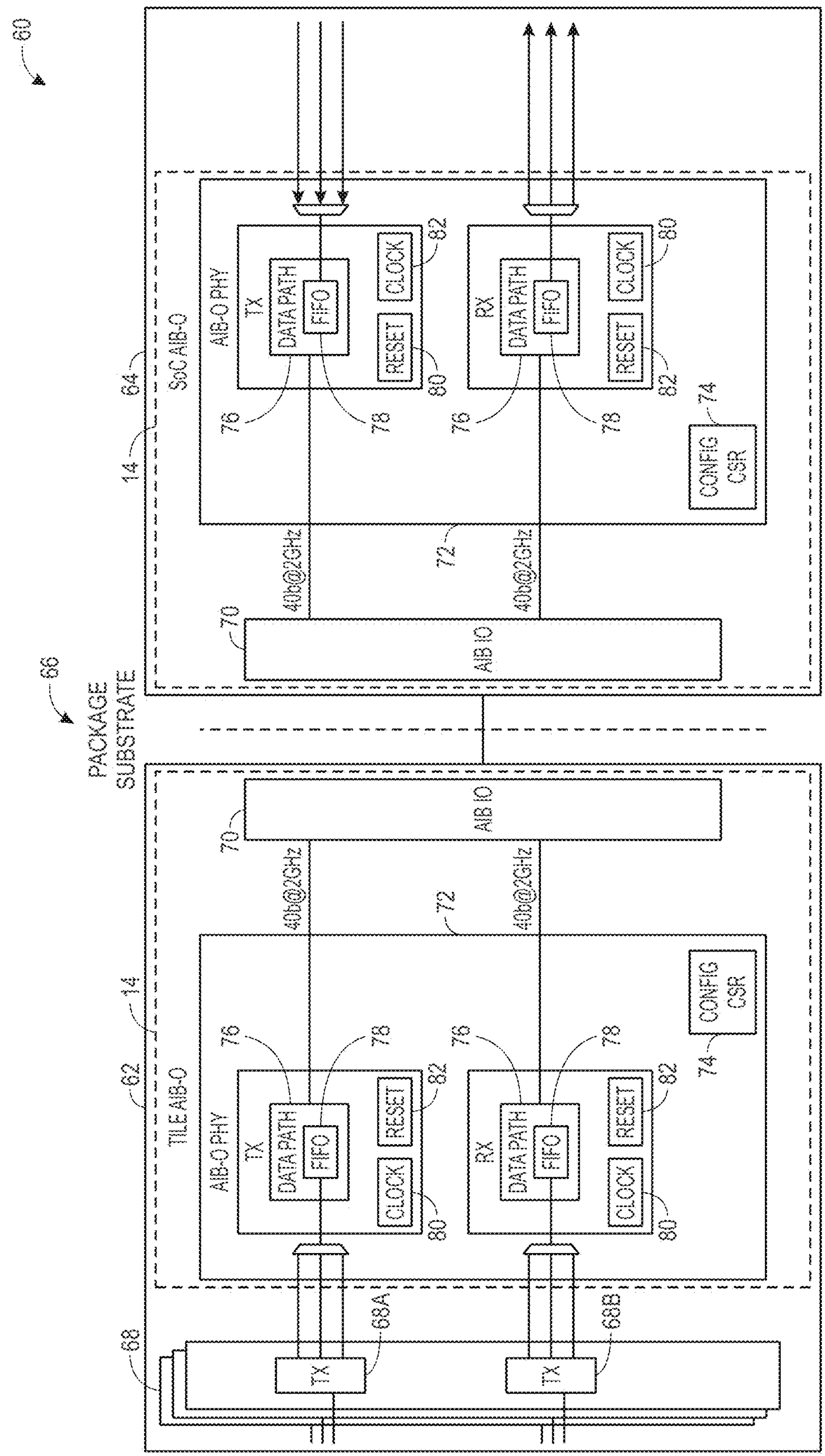
60
66
PACKAGE SUBSTRATE
14
64
SoC AIB-O
AIB-O PHY
TX
DATA PATH
FIFO
RESET
CLOCK
RX
CONFIG CSR
76
78
80
82
74
40b@2GHz
72
70
AIB IO
62
TILE AIB-O
68
68A
68B

FIG. 5

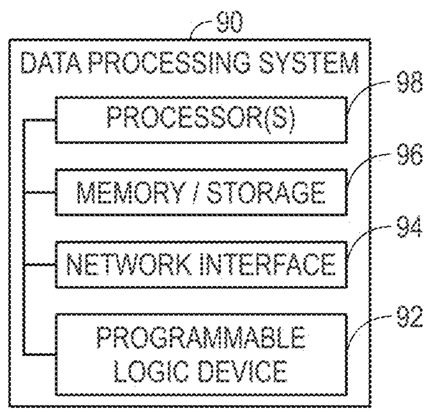
90
DATA PROCESSING SYSTEM
98
PROCESSOR(S)
96
MEMORY / STORAGE
94
NETWORK INTERFACE
92
PROGRAMMABLE
LOGIC DEVICE

*FIG. 6*

# SELECTIVE USE OF DIFFERENT ADVANCED INTERFACE BUS WITH ELECTRONIC CHIPS

## CROSS-REFERENCE TO RELATED APPLICATION

**[0001]** This a divisional application claiming priority to U.S. patent application Ser. No. 17/033,655, entitled "Selective use of Different Advanced Interface Bus with Electronic Chips," filed on Sep. 25, 2020, which is incorporated by reference in its entirety for all purposes.

## BACKGROUND

**[0002]** The present disclosure relates generally to electronic devices using multiple communicative electronic chips or chiplets.

**[0003]** Electronic systems and electronic devices are becoming more versatile and efficient in data processing to keep up with the ever increasing push for faster processing of data. Some data processing systems may include electronic devices that include multiple electronic chips and chiplets communicatively coupled to perform data processing tasks. The multiple chips or chiplets in a data processing task may be programmable logic devices, application-specific integrated circuits, processors, transceivers, or any other electronic circuit component capable of digital communication.

**[0004]** The aforementioned digitally communicative circuit components may use different interfaces for connection and communication with other circuit components. Such digitally communicative circuit components may use different sets of interconnect schemes with the respective interfaces for transmission and reception of data. The use of the various interconnect schemes may be due to various sets of communication rules imposed by a protocol, such as different speed rates, voltage levels, data encoding and decoding methods, and physical layouts, among other things. However, there is a need for more versatility with the interfaces of the digitally communicative circuit components with different communication capabilities. For example, different chips or chiplets of the digitally communicative circuit components may communicate using different numbers of input/output (I/O) ports.

**[0005]** The use of a specific interface in a specific digital component (e.g., chip or chiplet) may be optimal for the chip or chiplet design. Some chiplets may communicate using a high data bandwidth interface while other chiplets may communicate using interfaces with lower data bandwidth. Moreover, a scaled data processing system may use various chip or chiplets each using different numbers of I/O ports. Digital communication using high data bandwidth communication interfaces for all chiplets of such a scaled data processing system may be costly and inefficient. For example, the scaled data processing system may include one chiplet with high bandwidth while other communicating chiplets include circuitry with lower data bandwidth.

**[0006]** This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0007]** Advantages of the disclosure may become apparent upon reading the following detailed description and upon reference to the drawings in which:

**[0008]** FIG. **1** depicts a chiplet with an advanced interface bus 2.0 (AIB 2.0) or an advanced interface bus-O (AIB-O), in accordance with an embodiment;

**[0009]** FIG. **2** depicts input/output (I/O) of an AIB 2.0 scheme and an AIB-O scheme, in accordance with an embodiment;

**[0010]** FIG. **3** depicts a perspective view of the AIB 2.0 and the AIB-O of FIG. **1**, in accordance with an embodiment;

**[0011]** FIG. **4**. depicts a first and a second chiplet communicatively coupled using the AIB-O scheme and AIB 2.0 scheme interface, in accordance with an embodiment;

**[0012]** FIG. **5** depicts a first and a second chiplet communicatively coupled using the AIB-O scheme interfaces, in accordance with an embodiment; and

**[0013]** FIG. **6** depicts a transmitter circuitry and receiver circuitry coupled between chiplets of the package of FIG. **1**, in accordance with an embodiment.

## DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

**[0014]** One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It may be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it may be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

**[0015]** When introducing elements of various embodiments of the present disclosure, the articles "a," "an," and "the" are intended to mean that there are one or more of the elements. The terms "comprising." "including," and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to "one embodiment" or "an embodiment" of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, the phrase A "based on" B is intended to mean that A is at least partially based on B.

**[0016]** Certain aspects commensurate in scope with the originally claimed disclosure are set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of certain forms of the disclosure might take and that these aspects are not intended to limit the scope of the disclosure. Indeed, the disclosure may encompass a variety of aspects that may not be set forth below.

**[0017]** Methods and systems for scaling die-to-die communication are described. Data processing system may benefit from additional versatility from a chiplet interface to facilitate communication between chiplets having different bandwidth capabilities. For example, an alternate interface may be used with communicative chiplets with multiple different data bandwidths to facilitate communication with other chiplets. Different embodiments are described below to facilitate a cost effective communication between chiplet with different data bandwidths. A chiplet may include at least a die, an interface layer, and packaging. The interface layer may facilitate digital communication between the die and other chiplets or chips. The interface layer may be implemented using a physical layer to provide high data bandwidth or a different physical layer to provide lower data bandwidth. A data bandwidth density of the physical layer may depend on a number and proximity of conductive traces within the physical layer. An application of the chiplet in a processing system (e.g., based on communication capability of other die within a system of chiplets or chips) may impose data bandwidth requirement for the interface layer. As such, the die may be packaged with an interface layer including a high-density physical layer to provide a high data bandwidth or a low-density physical layer to provide a lower data bandwidth.

**[0018]** Packaging the same die with different interface layers with different data bandwidth may facilitate versatile and scalable processing system design. However, such interface layers may use the same communication protocols. That is, an interconnection scheme associated with physical layers of such interface layers with different data bandwidth may facilitate data communication with similar protocols.

**[0019]** That said, a system of chips or chiplets may include multiple standardized chiplet interfaces to facilitate scalable system design. For example, a processing system may use a high data bandwidth density interface technology (e.g., an embedded multi-die interconnect bridges (EMIB)) with a low data bandwidth density interface technology. As such, a complexity of system design may be reduced using different interface layers with high data bandwidth and lower data bandwidth while using the same communication protocol. Methods and systems for using die-to-die interface scaling for high interconnect density EMIB/interposer to low-interconnect-density package traces are described.

**[0020]** Assisting packaging technology for implementing high data bandwidth density chiplets with organic substrates may be used for a variety of purposes. Some chiplet interfaces may be directed to advanced packaging. For example, in different embodiments, a chiplet may use an Embedded Multi-die Interconnect Bridge (EMIB), a silicon interposer, or active interposer, among other chiplet interface technologies. However, such advanced packaging technologies may have more associated costs for providing high functionality that may not be needed/usable for some applications. For example, some analog to digital convertors (ADCs), digital to analog convertors (DACs), and transceivers, among other things, may only require a fraction of data bandwidth density provided by the chiplet interfaces with such advanced packaging technologies.

**[0021]** A scalable packaging technology may be useful to provide sufficient functionality commensurate with design requirements in different chiplet or chip applications. Such a scalable packaging technology may provide cost effective a communication solution between a system of chiplets while preserving functionality of chiplets and providing cost effective solution for system design implementation.

**[0022]** As will be appreciated, different chiplet interface (or interconnect) layers may use different manufacturing processes and/or different materials. An interface layer may include a redistribution layer (RDL), a number of bumps, and a substrate that includes conductive traces to enable external communication with other chiplets or chips. As such, a cost associated with implementation of a respective chiplet, chip, or system of chips may vary based on the interface layers used with chiplets to facilitate data communication. For example, some chiplets may use an Advanced Interface Bus 1.0 (AIB 1.0) interconnect implemented on a substrate made of organic material. A relatively small concentration of traces within the substrate and a relatively small bandwidth density implemented on the RDL and the bumps of the interface layer provides for the use of organic material with the AIB 1.0. Moreover, AIB 1.0 may facilitate communication using a first communication protocol.

**[0023]** On the other hand, Advanced Interface Bus 2.0 (AIB 2.0) may include high data bandwidth with concentrated RDL and bumps in the respective interface layer (high density) and highly concentrated traces within the respective substrate. However, the highly concentrated interface layer uses denser and finer connections within the RDL, finer bump sizes, and more sophisticated substrate material to facilitate finer interface and interconnection with other chiplets. As such, AIB 2.0 may be implemented using more expensive substrate or bump material, such as silicon. Moreover, AIB 2.0 may facilitate communication using a second communication protocol different from the first communication protocol.

**[0024]** Some advanced chiplets may communicate with other advanced chiplets with high data bandwidth within a communication system. Such communication system may benefit from using Advanced Interface Bus 2.0 (AIB 2.0) interconnect technology. However, some other communication systems may use communications between an advanced chiplet communicating (using the AIB 2.0) with another chiplet with lower data bandwidth density (using the AIB 1.0). Such communication may not be viable due to the use of different communication protocol by AIB 1.0 and AIB 2.0. AIB 2.0 may facilitate communication between the advanced chiplet and the other chiplet. However, using AIB 2.0 for chiplets that are capable of performing adequately using the lower bandwidth interface increases a cost of the system because of high density interface incorporated using a more expensive substrate (e.g., silicon).

**[0025]** AIB-O is described hereinafter for facilitating communication using a communication protocol usable with (and similar to) the AIB 2.0 protocol, while including less dense conductors in an interface layer of the chiplet (including the RDL and bumps) and less density in substrate. AIB-O may facilitate communication between the advanced chiplet and the other chiplet (with less data bandwidth) using organic materials for the interface layer and substrate to lower the cost. The AIB-O may include a fraction of bumps used with AIB 2.0 while using the similar communication protocols (e.g., interconnect scheme) on similar footprints.

**[0026]** Referring now to FIG. **1**, a chiplet **10** may include an AIB 2.0 layer **12** or an AIB-O layer **14**. Although the following discusses AIB 2.0 and AIB-O, similar techniques may be applied to similar high-density and low-density protocols and/or layers. Incorporating the AIB 2.0 layer **12** or the AIB-O layer **14** with the chiplet **10** may be selectable at the time of manufacturing the chiplet **10**, as will be appreciated. The chiplet **10** may include a respective die **16**. The die **16** may include multiple layers of processing elements, such as a logic layer or a transistor layer, and may communicate using different data bandwidth. For example, the die **16** may be facilitate a field programmable gate array (FPGA), digital signal processor (DSP), an application specific integrated circuit (ASIC), hard logic, soft logic, among other things. Nevertheless, the die **16** may communicate with one or more other chiplets via an interface layer, such as the AIB 2.0 layer **12** or AIB-O layer **14**, incorporated on a selectable layer **18**.

**[0027]** The chiplet **10** may communicate with other die, chiplets, or chips using the selectable layer **18** that may incorporate the interface layer. In some embodiments, the selectable layer **18** may include one or more top metal layers of the chiplet **10** to facilitate interconnection scheme between the die **16** and external circuitry. As such, the chiplet **10** may include the die **16** packaged with the selectable layer **18** to provide data communication using a data communication protocol. The data communication protocol may include a specific data rate, frequency, bandwidth density, etc. In different embodiments, the selectable layer **18** may be manufactured using different material (e.g., silicon, organic materials) to facilitate data communication with the other die using the respective interface layer (e.g., AIB 2.0 layer **12** or the AIB-O layer **14**).

**[0028]** The selectable layer **18** may include an RDL layer and bumps, among other things. The RDL layer and bumps may facilitate data communication between the die **16** and other chiplets via a substrate. In specific embodiments, to facilitate digital communication with other chiplets, the die **16** may use the same communication protocol when the chiplet **10** is using the AIB 2.0 layer **12** or the AIB-O layer **14**. Moreover, because the AIB 2.0 layer **12** or the AIB-O layer **14** may facilitate communication using the same communication protocol, the die **16** may communicate with other chiplets with an AIB 2.0 scheme or an AIB-O scheme.

**[0029]** The AIB-O layer **14** may include a fraction (e.g., ½, ⅓, ¼, etc.) of data bandwidth as compared to AIB 2.0 layer **12** using the same footprint while using the same communication protocol. For example, the die **16** may be communicatively coupled to another chiplet with high data bandwidth. As such, the die **16** may be implemented using the AIB 2.0 layer **12** to facilitate the high data bandwidth. However, in another example, the die **16** may be communicatively coupled to a different chiplet with relatively lower data bandwidth density. As such, the die **16** may be implemented using the AIB-O layer **14** to facilitate the relatively lower data bandwidth. The die **16** may communicate with the other chiplets using the same communication protocol in both examples. In specific embodiments, the AIB-O layer **14** may include a fraction (e.g., ½, ⅓, ¼, etc.) of the bumps used with a AIB 2.0 scheme on the same footprint (e.g., selectable metal layer **18**). As such, a bandwidth density of a physical layer of the AIB-O layer **12** (including the RDL and the bumps) may be a fraction of a bandwidth density of a physical layer of the AIB 2.0 layer **14**. In other embodiments, the AIB-O layer **14** may be parametrized and a manufacturer may provide the chiplet **10** with AIB-O layer **14** using different ratios compared to AIB 2.0 layer **12** (e.g., 8:1, 4:1, 2:1 and 1:1)

**[0030]** However, the selectable layer **18** may include different physical properties when using the AIB 2.0 layer **12** or the AIB-O layer **14**. That is, because the AIB-O layer **14** includes a fraction of bumps compared to AIB 2.0 layer **12** on the same footprint, the chiplet **10** incorporated with the AIB-O layer **14** may be manufactured using a wider variety of materials. For example, the AIB-O layer **14** may include organic material for RDL layer and bumps, whereas, the AIB 2.0 layer **12** may include silicon with RDL layer and bumps to incorporate the high density layer with finer design and implementation. It should be appreciated that in specific embodiments, a die **16** may communicate using the AIB 2.0 layer **12**, the AIB-O layer **14**, or a combination of the two interface layers on different portion (e.g., different I/O banks) of the same footprint (e.g., selectable metal layer **18**) to facilitate data communication.

**[0031]** In a first embodiment, the chiplet **10** may use the die **16** with the AIB 2.0 layer **12** to communicate with one or more other chiplets with high data bandwidth (e.g., 256 Gbps). The AIB 2.0 layer **12** may use an advanced circuitry material, including high data bandwidth density in the physical layer, to facilitate such data communication. For example, the AIB 2.0 layer **12** may use Micro-bumps (μbumps) implemented using silicon substrate material. In specific embodiments, the μbumps may use 45 millimeter (mm) or another measurement for pitch size. Moreover, the μbumps or the silicon substrate may increase the manufacturing cost associated with the chiplet **10**. However, some applications of the die **16** may have sufficient bandwidth with a lower data bandwidth density while benefitting from other functionality associated with the die **16**. For example, the chiplet **10** may communicate with one or more other chiplets with lower data bandwidth density capability than AIB 2.0 specifications.

**[0032]** In a second embodiment, the chiplet **10** may use the die **16** with the AIB-O layer **14** to communicate with one or more other chiplets using lower data bandwidth (e.g., 56 Gbps). The second embodiment may be used in different applications than the first embodiment. Specifically, the second embodiment may be used when a communication with lower data bandwidth capability is sufficient for communication with the one or more other chiplets. The AIB-O layer **14** may provide an interface, using lower data bandwidth density in a respective physical layer, between the chiplet **10** (e.g., high cost die or chiplet) and a lower-cost die or chiplets using lower data bandwidth density in the physical layer (e.g., EDL layer and bumps).

**[0033]** As previously noted, the AIB-O layer **14** may provide a fraction of data communication bandwidth compared to AIB 2.0 layer while preserving other functionality of the die **16** (e.g., same communication protocol). The AIB-O layer **14** may also include a fraction number of bumps used in the top layer of the chiplet **10** compared to the AIB 2.0 layer **12**. In specific embodiments, the AIB-O layer **14** may facilitate data communication using C4 bumps and the AIB 2.0 layer **12** may facilitate data communication using μbumps. The C4 bumps may be wider and may occupy more space on a top layer of the chiplet **10** compared to μbumps. An implementation of the C4 bumps for die to die communication interfaces may allow the use of more variety of materials for manufacturing the bumps compared to μbumps. For example, a die may use the AIB-O layer **14** to communicate with other die through organic material substrates. As such, the AIB-O layer **14** may provide scalable communication with other chiplets that use lower data communication bandwidth densities while preserving other functionality (e.g., frequency, data rate, etc.) of the die **16**.

**[0034]** The use of AIB-O layer **14** with the die **16** may reduce associated manufacturing costs while enabling scalable designs by reducing an edge data bandwidth per unit perimeter. As such, the AIB-O layer **14** and the AIB 2.0 layer **12** may facilitate scalable system designs by accounting for a tradeoff between data bandwidth density for die-to-die communication and implementation cost by providing different chiplets using the same die **16**.

**[0035]** In some embodiments, an FPGA may be implemented using the AIB-O layer **14**. Depending on an application and a bandwidth density for the application, the cost of system implementation may be reduced by replacing AIB 2.0 layer **12** with AIB-O **14** using the same FPGA. That is, the FPGA may be implemented using a first metal layer with the AIB-O layer **14** or a second metal layer using the AIB 2.0 layer **12** (with different packaging technologies). As a result, two different versions of the FPGA may be manufactured using the same die. In other embodiments, an ASIC or other processing blocks may use the AIB-O layer **14**.

**[0036]** With the foregoing in mind, FIG. **2** depicts I/O ports **20** associated with the AIB 2.0 layer **12** and I/O ports **22** associated with the AIB-O layer **14**. A processing system such as a system on a chip, or other computing component may include the chiplet **10**. A first embodiment of the chiplet **10** may use the AIB 2.0 layer **12** with the AIB 2.0 I/O ports **20** and a second embodiment of the chiplet **10** may use the AIB-O layer **14** with the AIB-O I/O ports **22**. The first and the second embodiments may use the same die (e.g., die **16** of FIG. **1**) with the chiplet **10**. As such, an interface layer (selectable layer **18** of FIG. **1**) may be implemented using the AIB 2.0 layer **12** including the AIB 2.0 I/O ports **20** or the AIB-O layer **14** including the AIB-O I/O ports **22** on the same footprint (top metal layer of the chiplet **10**). In the specific example, the AIB 2.0 I/O ports **20** may be implemented using μbumps and the AIB-O I/O ports **22** may be implemented using C4 bumps.

**[0037]** In FIG. **2**, the AIB-O I/O ports **22** may include a fraction of the number of I/O as the AIB 2.0 I/O ports **20** on the same footprint, as will be appreciated. For example, the μbumps may be implemented using a finer pitch size while C4 bumps may be implemented using wider pitch size. As such, the smaller number of I/O ports associated with AIB-O I/O ports **22** may facilitate lower data bandwidth density (per perimeter) on the chiplet **10**. For example, an AIB-O scheme, including the AIB-O I/O ports **22** (using C4 bumps), may facilitate data communication using a fraction of data bandwidth as compared to an AIB 2.0 scheme, including the AIB 2.0 I/O ports **20** (using μbumps), while facilitating communication of a die (e.g., the die **16** of FIG. **1**) with other chiplets using the same communication protocol as AIB 2.0 scheme. As such, while the AIB-O scheme may include a physical layer (including the AIB-O I/O ports **22**) with a lower bandwidth density, the interface between the die and other chiplets (using AIB 2.0 scheme or AIB-O scheme) may remain consistent.

**[0038]** In specific embodiments with respect to FIG. **2**, the AIB-O I/O ports **22** may include a quarter of the number of I/O ports as compared to the AIB 2.0 I/O ports **20**. For example, each C4 bump (used to facilitate AIB-O I/O ports **22**) may be wider and may occupy more space as compared to a μbump (used to facilitate AIB 2.0 I/O ports **20**). In these embodiments, less data communication routes may be used/assigned between the die and other communicative chiplets. For example, the AIB-O layer **14** may use a specific RDL with AIB-O I/O ports **22** that reconnects the die I/O to C4 bumps of the chiplet **10** to facilitate data communication. Moreover, a consistent communication protocol may facilitate data transmission and reception using the die of chiplet **10** and other chiplets using the AIB 2.0 I/O ports **20** with an AIB 2.0 scheme or the AIB-O I/O ports **22** with an AIB-O scheme to facilitate scalable design in a system of chiplets.

**[0039]** As described above, the AIB-O I/O ports **22** may facilitate data communication using a fraction of a full-bandwidth density of the chiplet **10** (and the respective die). In specific embodiments, the AIB-O I/O ports **22** may facilitate using the fraction of the full-bandwidth density of the chiplet **10** to provide design scalability to system design using chiplet **10**. For example, the die may communicate with one or more other chiplets that require a fracture of the full-bandwidth density of the die. Moreover, such communication may be implemented using only one or a number of channels of the chiplet **10**. An RDL layer of the chiplet **10** associated with the AIB-O layer **14** may facilitate routing of communications between the die and the AIB-O I/O ports **22**.

**[0040]** The AIB 2.0 I/O ports **20** may include a power I/O portion **24**A, a power I/O portion **24**B, and a communication I/O portion **26**, among other portions. The power I/O portion **24**A and the power I/O portion **24**B may include I/O ports with input power pins or output power pins for the chiplet **10**. The communication I/O portion **26** may include I/O ports for data communication via the die of the chiplet **10**. In some embodiments, the power I/O portion **24**A, the power I/O portion **24**B, and the communication I/O portion **26** may be user configurable. Moreover, in different embodiments, the AIB 2.0 I/O **20** may include different I/O portions or combination of I/O portions.

**[0041]** The AIB-O I/O ports **22** may include a power I/O portion **28**A, a power I/O portion **28***b*, and a communication I/O portion **30**. The power I/O portion **28**A and the power I/O portion **28**B may be associated with input power pins or output power pins of the chiplet **10**. The communication I/O portion **26** may include I/O ports for data communication via the die of the chiplet **10**. In some embodiments, the power I/O portion **28**A, the power I/O portion **28***b*, and the communication I/O portion **30** may be user configurable. Moreover, in different embodiments, the AIB-O I/O ports **22** may include different I/O portions or combination of I/O portions.

**[0042]** As depicted in FIG. **2** and described above, the AIB-O I/O ports **22**, including the power I/O **28**A, the power I/O **28***b*, and the communication I/O **30**, may include the fraction of the number of I/O ports as compared to the AIB 2.0 I/O ports **20**. The AIB-O layer **14** may be implemented using a wider variety of manufacturing material, such as organic materials. As such, the AIB-O layer **14** may facilitate scalable and cost-effective system designs for communicating with other chiplets using the AIB-O I/O ports **22** (with an AIB-O scheme) or AIB 2.0 I/O ports **20** (with an AIB 2.0 scheme).

**[0043]** Referring now to FIG. **3**, I/O pins associated with the AIB 2.0 layer **12** and the AIB-O layer **14** are overlayed on the top metal layer of the chiplet **10** in a perspective view for comparison. As mentioned above, in different embodiments, the one or more top layers of a die, such as the die **16** of FIG. **1**, of the chiplet **10** may use one or the other interface bus technologies (AIB 2.0 or AIB-O) for data communication with other die. The overlayed perspective view of FIG. **3** provides an illustrative comparison between the respective I/O pins of the AIB 2.0 layer **12** and the AIB-O layer **14**. It should be noted that FIG. **3** may depict only a portion of the chiplet **10**.

**[0044]** In FIG. **3**, a number of I/O pins associated with the AIB 2.0 layer **12** are implemented using μbumps and other I/O pins associated with the AIB-O layer **14** are implemented using C4 bumps. In some embodiments, the μbumps may use pitch bumps having a narrower pitch (e.g., 45 mm), whereas the C4 bumps may use relatively wider pitch bumps. As such, the number of C4 bumps disposable on the AIB-O layer **12** may be a fraction of the μbumps disposable on the AIB 2.0 layer **12**. In specific embodiments, the μbumps may be implemented using a silicon substrate for implementing high data bandwidth density in a more sophisticated design. The C4 bumps may be implemented using a substrate with a wider variety of substrate materials that may enable using the die in applications when lower data bandwidth is sufficient. For example, the C4 bumps may be implemented using organic materials because of the smaller bandwidth density used in the physical layer. As described above, the μbumps may be associated with the AIB 2.0 layer and used with EMIB technology while the C4 bumps may be used to implement the AIB-O layer **14** with AIB-O scheme.

**[0045]** Each depicted I/O bank may include a number of I/O pins to facilitate data communication between the chiplet **10** and other chiplets. In some embodiments, each I/O bank may be associated with one or more functional blocks of the die determined at the time of manufacturing. Alternatively, the I/O banks may be user configurable after the manufacturing process. Nevertheless, in different embodiments, each I/O bank may be associated with a number of the μbumps or the C4 bumps. As described above and depicted in the embodiment of FIG. **3**, each I/O bank may use a first number of pins when using μbumps with the AIB 2.0 layer **12**, or a fraction of the first number of pins when using the C4 bumps with the AIB-O layer **14**. The different number of bumps is partly due to different pitch sizes used with different bump technologies. As such, each I/O bank may communicate using a physical interface layer with a higher data bandwidth density when using the μbumps compared to C4 bumps (the AIB 2.0 layer **12** and the AIB-O layer **14**).

**[0046]** It should be noted that the use of μbumps and C4 bumps are with respect to specific embodiments and other embodiments may use other bumps with the chiplet **10**. For example, in the AIB-O scheme, the chiplet **10** may use a fraction (e.g., a quarter) of the bumps (or I/O pins) compared to AIB 2.0 scheme which allows the AIB-O layer **14** to use various bump technologies because of lower design restrictions. That is, AIB-O layer **14** is not limited to C4 bumps.

**[0047]** Moreover, the top metal layer of the chiplet **10** may include one or more layers or sub-layers for routing of the I/O pins to the respective ports on the die. For example, when using the AIB 2.0 layer **12**, the chiplet **10** may include an RDL for directing the μbumps to the respective sections/ports of the die. Similarly, when using the AIB-O layer **14**, the chiplet **10** may include another RDL for connecting the C4 bumps to the respective parts of the die.

**[0048]** With the foregoing in mind, in some embodiments, the AIB-O layer **14** may use an RDL on the top metal layer of the chiplet **10** similar to that of AIB 2.0 layer **12**. In such embodiments, one or more I/O pins of the die may not be used since AIB-O layer **14** may use a fraction of bumps compared to AIB 2.0 layer **12**. In other embodiments, an RDL associated with the AIB-O layer **14** may include a different routing circuitry for making the AIB-O layer **14** bumps compatible with the die I/O ports. As such, the RDL associated with the AIB-O layer **14** may facilitate the reduced data bandwidth density of the die **16** on the same footprint by not using a number of the die I/O ports or rewiring the die I/O ports.

**[0049]** A manufacturer may use the AIB 2.0 layer **12**, the AIB-O layer **14**, or a combination of the two (e.g., on different I/O banks) to implement the top layers of the chiplet **10**. In some embodiments, the chiplet **10** may include different interfaces on different channels or different sides of the package. In some other embodiments, a first chiplet may include one or more interfaces using the AIB 2.0 layers **12** and a second chiplet may include one or more interfaces using the AIB-O layer **14**. A die may use the AIB 2.0 layer **12** for communication with advanced chiplets using high data bandwidth or use the AIB-O layer **14** for communication with chiplets with lower data bandwidth. The die may include the same processing circuitry in either embodiments (e.g., logic circuitry, transistor layer, etc.) connected to AIB 2.0 layer **12** or AIB-O layer **14**. AIB-O scheme implemented by the AIB-O layer **14** may facilitate similar functionality and reduced data bandwidth density when the die is in communication with other die with lower data bandwidth while using the same protocol as the AIB 2.0 scheme. As such, system designs may benefit from scalable designs using the AIB-O layer **14** and the AIB 2.0 layer **12**.

**[0050]** FIG. **4**. depicts an example communication system **40**. The communication system **40** may include a first chiplet **42** and a second chiplet **44**. The first chiplet **42** and the second chiplet **44** may be communicatively coupled via a substrate **46**. The first chiplet **42** and the second chiplet **44** may each include one of an FPGA, Application Specific Integrated Circuit (ASIC), and the like. The first chiplet **42** may communicatively couple to the substrate **46** using the AIB 2.0 layer **12** and the second chiplet **44** may communicatively couple to the substrate **46** using the AIB-O layer **14**.

**[0051]** The AIB 2.0 layer **12** associated with the first chip **42** may include channels **48**. The channels **48**A to **48**G may each communicate as a transmitter or receiver channel. For example, the channels **48**A, **48**C, **48**E, and **48**G may be configured to transmit communication data to the second chiplet **44**, whereas, the channels **48**B, **48**D, and **48**F may be configured to receive communication data from the second chiplet **44**.

**[0052]** Similarly, the AIB-O layer **14** associated with the second chip **44** may include channels **50**. The channels **50**A to **50**G may each communicate as a transmitter or receiver channel. For example, the channels **50**A, **50**C, **50**E, and **50**G may be configured to transmit communication data to the first chiplet **42**, whereas, the channels **50**B, **50**D, and **50**F may be configured to receive communication data from the first chiplet **42**. It should be noted that in other embodiments, the first chiplet **42** and the second chiplet **44** may include different numbers of channels and may communicate using different configurations of channels (e.g., transmitter and receiver channels).

**[0053]** In some embodiments, a sideband **52**, associated with the first chiplet **42**, and a sideband **44**, associated with the second chiplet **44** may include channel configuration data. For example, the sideband **52** may store configuration data associated with an architecture of channels **48** and the sideband **52** may store configuration data associated with an architecture of channels **50**. The sideband **52** and the sideband **54** may transmit the respective configuration data to configure the respective AIB 2.0 layer **12** and the AIB-O layer **14** to facilitate assignment of the respective I/O pins.

**[0054]** The AIB 2.0 layer **12** may include high data bandwidth density on physical layers associated with the channels **50**A to **50**G. The AIB 2.0 layer **12** may include μbumps. The AIB-O layer **14** may include smaller data bandwidth density on physical layers associated with the channels **50**A to **50**G. The AIB-O layer **14** may include C4 bumps. As previously noted, the C4 bumps associated with the AIB-O layer **14** may be a fraction of the μbumps associated with the AIB 2.0 bumps. However, the first chiplet **42** and the second chiplet **44** may communicate using the same communication protocol without a translation circuitry between the AIB 2.0 layer **12** and the AIB-O layer **14**. As such, the first chiplet **40** and the second chiplet **42** may communicate using the same clocking, initialization, self-tuning and sideband architecture that will scale across varying data widths to facilitate the use of the same eco-system across multiple physical technologies.

**[0055]** The AIB-O layer **14** may have different physical characteristics compared to the AIB 2.0 layer. Due to the smaller number of bumps, the second chiplet **44** may use smaller bandwidth density for the respective interface layer. Moreover, the substrate **46** may include smaller number of traces with less physical density. For example, the smaller number of bumps of the AIB-O layer **14** may limit a number of traces and the density and concentration of the traces withing the substrate **46** for data communication between the first chiplet **42** and the second chiplet **44**. As such, a wider variety of manufacturing material (e.g., cost-effective material) may be used for the substrate **46**. However, this AIB-O layer **14** may communicate using similar clocking, initialization, self-tuning and sideband architectures. A such, the AIB-O layer **12** may scale across different chiplets with different data widths to facilitate an eco-system of chiplets across multiple physical technologies with different bandwidth density.

**[0056]** Referring now to FIG. **5**, a first chiplet **62** and a second chiplet **64** may communicate using AIB-O layers **14** in a communication system **60**. In different embodiments, the first chiplet **62** or the second chiplet **64** may be an FPGA. ASIC, or other processing component. A substrate **66**, similar to substrate **46** of FIG. **4**, may facilitate communication between the first chiplet **62** and the second chiplet **64**. For example, the first chiplet **62** and the second chiplet **64** may each include a number (e.g., 22) C4 bumps (not shown in FIG. **5**) to transfer data at a corresponding rate (e.g., 4 Gbps).

**[0057]** In the depicted embodiment, the first chiplet **62** may include communication channels **68** including a transmission channel **68**A and a receiver channel **68**B. The communication channels **68** may provide the data for transmission using the AIB-O layer **14**. The AIB-O layer **14** may facilitate data communication using AIB-O scheme in each of the first chiplet **62** and the second chiplet **64**. An example embodiment of circuitry associated with the AIB-O layer **14** is described with respect to communication system of FIG. **5**.

**[0058]** Each of the AIB-O layers **14** may include an AIB-O I/O block **70** and an AIB-O physical layer **72**. The AIB-O I/O block **70** may include C4 bumps as described with respect to FIG. **3** or other viable bump technologies. The AIB-O I/O block **70** may include bump architectures with respect to AIB-O I/O ports **22** described with respect to FIG. **2**. The AIB-O physical layer **72** may be associated with RDL and provide processing and routing circuitry to facilitate data communication. In one embodiment, the AIB-O physical layer **72** may facilitate data communication using the functional blocks described herein and with respect to FIG. **5**

**[0059]** The AIB-O physical layer **72** may include a configuration data **74**, a data path block **76**, a first-in-first-out register (FIFO) **78**, a clock **80**, and a reset **82**. However, it should be appreciated that in different embodiments, the AIB-O physical layer **72** may include different circuitry and circuit blocks. The configuration data **74** may include data indicative of architecture of the AIB-O physical layer **72**. For example, a sideband, such as sideband **54** of FIG. **4** may provide the configuration data **74**. The data path block **76** may include the FIFO **78**, clock **80**, and reset **82** to facilitate routing of the data between the communication channels **68** and the AIB-O I/O block **70**. In some embodiments, a respective RDL layer may incorporate the data path block **76**. In different embodiments, the FIFO **78** may receive and transmit data for transmission or reception. The clock **80** may facilitate a rate at which the FIFO **78** may transmit data. The reset **82** may be triggered by a control signal to reconfigure the data path block **76**. For example, the configuration data **74** may provide such trigger signal.

**[0060]** A data processing system **90** may include one or more programmable logic devices **92** incorporating the flexible interface techniques discussed herein, shown in FIG. **6**. The data processing system **90** includes a processor **98**, memory and/or storage circuitry **96**, and a network interface **94**. The data processing system **90** may present embodiments that may utilize the AIB-O scheme (i.e., AIB-O layer **14**). The data processing system **90** may include more or fewer components (e.g., electronic display, user interface structures, application specific integrated circuits (ASICs)). The processor **98** may include any suitable processor, such as an INTEL® XEON® processor or a reduced-instruction processor (e.g., a reduced instruction set computer (RISC), an Advanced RISC Machine (ARM) processor) that may manage a data processing request for the data processing system **90** (e.g., to perform machine learning, video processing, voice recognition, image recognition, data compression, database search ranking, bioinformatics, network security pattern identification, spatial navigation, or the like). The memory and/or storage circuitry **96** may include random-access memory (RAM), read-only memory (ROM), one or more hard drives, flash memory, or the like. The memory and/or storage circuitry **96** may be considered external memory to the programmable logic device **92** and may hold data to be processed by the data processing system **90**. In some cases, the memory and/or storage circuitry **96** may also store configuration programs (e.g., bitstream) for programming the programmable logic device **92**. The network interface **94** may enable the data processing system **90**

to communicate with other electronic devices and may perform the communication by the way of the AIB-O scheme (using the AIB-O layer **14**) or the AIB 2.0 scheme (e.g., using the AIB 2.0 layer **12**). Furthermore, the processor **98**, the programmable logic device **92** and/or the memory and/or storage circuitry **96** may utilize different communication features and may benefit from the AIB-O layer or AIB 2.0 layer. The use of AIB-O layer or AIB 2.0 layer in between the functional blocks of the data processing system **90** may enable scalable design using the functional blocks and may reduce a cost of circuit implementation, among other things.

**[0061]** In one example, the data processing system **90** may be part of a data center that processes a variety of different requests. For instance, the data processing system **90** may receive a data processing request via the network interface **94** to perform machine learning, video processing, voice recognition, image recognition, data compression, database search ranking, bioinformatics, network security pattern identification, spatial navigation, or some other specialized task. The processor **98** may cause the programmable logic fabric of the programmable logic device **92** to be programmed with a particular accelerator related to requested task. For instance, the processor **98** may instruct that configuration data (bitstream) stored on the memory and/or storage circuitry **96** or cached in sector-aligned memory of the programmable logic device **92** to be programmed into the programmable logic fabric of the programmable logic device **92**. The configuration data (bitstream) may represent a circuit design for a particular accelerator function relevant to the requested task. Due to the high density of the programmable logic fabric, the proximity of the substantial amount of sector-aligned memory to the programmable logic fabric, or other features of the programmable logic device **92** that are described here, the programmable logic device **92** may rapidly assist the data processing system **90** in performing the requested task. Indeed, in one example, an accelerator may assist with a voice recognition task less than a few milliseconds (e.g., on the order of microseconds) by rapidly accessing and processing large amounts of data in the accelerator using sector-aligned memory.

**[0062]** Placement of computation and memory in spatial architectures where compute and memory have three-dimensional spatial locality may be performed statically. Additionally or alternatively, the programmable logic device **92** may dynamically allocate, relocate, and de-allocate compute and memory on such spatial architectures. These techniques enable the static mapping and dynamic management of systems using such architectures. Moreover, using flexible allocation schemes enables the programmable logic device **92** to find and support optimal co-placement of compute with memory in a static setting in a known sequence of static settings and in a dynamic setting when the allocation of compute and memory is not known a priori. Such usage of static and/or dynamic placement in a three-dimensional spatial locality provides the ability to extend compilation to support the simultaneous synthesis, placement, and routing of spatial computation with a spatially distributed memory to enable users to leverage an architecture with a much richer memory sub-system. The support for dynamic management of the computation and memory allows users/administrators to build dynamic runtime systems for spatial architectures for the programmable logic device **92**.

**[0063]** Furthermore, the AIB-O layer **14** eliminates translation using translation circuitry between certain communicative components to provide flexibility of using different technologies for the data processing system **90**. For instance, the data processing system **90** may use the AIB-O layer **14** to readily communicate using the same protocols with different interface hard layers for a reduced time-to-market for each sub-component. For instance, new technologies may be used with other technologies (e.g., medium-dependent interface (MDI)) without redesigning pre-existing devices using the other technologies or designing the new technologies with different interface compatibility incorporated therein. Such compatibility may enable consistent communication between processors **98** (e.g., INTEL XEON®) and/or programmable logic devices **92** while reducing system implementation costs for the data processing system **90** compared to alternative tile interconnect approaches.

**[0064]** The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as "means for [perform]ing [a function] . . . " or "step for [perform]ing [a function] . . . ", it is intended that such elements are to be interpreted under 35 U.S.C. 112 (f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112 (f).

Example Embodiments

**[0065]** EXAMPLE EMBODIMENT 1. A semiconductor device comprising:

**[0066]** an interface comprising:

**[0067]** a physical layout of circuitry configured to be compatible with a high-speed protocol between die; and

**[0068]** a plurality of bumps having a density less dense than specified for the high-speed protocol and suitable for a low-speed protocol, wherein the interface is configured to communicate using the high-speed protocol.

**[0069]** EXAMPLE EMBODIMENT 2. The semiconductor device of example embodiment 1, comprising one or more die communicatively coupled to the die-to-die interface via a redistribution layer.

**[0070]** EXAMPLE EMBODIMENT 3. The semiconductor device of example embodiment 1, wherein a first die is communicatively coupled to a second chiplet via the interface.

**[0071]** EXAMPLE EMBODIMENT 4. The semiconductor device of example embodiment 3, wherein the first die and the second chiplet are configured to communicate using an advance interface bus (AIB) 2.0 protocol.

**[0072]** EXAMPLE EMBODIMENT 5. The semiconductor device of example embodiment 4, wherein the interface is configured to have a throughput that is a fraction of the AIB 2.0 protocol throughput.

**[0073]** EXAMPLE EMBODIMENT 6. The semiconductor device of example embodiment 5, wherein the throughput is equal to or less than 64 Gigabits per second.

[0074] EXAMPLE EMBODIMENT 7. The semiconductor device of example embodiment 5, wherein the throughput is of the interface is less than or equal to one quarter specified for AIB 2.0.

[0075] EXAMPLE EMBODIMENT 8. The semiconductor device of example embodiment 5, wherein the throughput is parametrized and configurable by a manufacturer of the semiconductor device.

[0076] EXAMPLE EMBODIMENT 9. The semiconductor device of example embodiment 1, comprising a substrate made of organic materials to which the plurality of bumps are coupled.

[0077] EXAMPLE EMBODIMENT 10. The semiconductor device of example embodiment 1, wherein the bumps have a pitch wider than 45 millimeter.

[0078] EXAMPLE EMBODIMENT 11. The semiconductor device of example embodiment 1, wherein the bumps comprise C4 bumps.

[0079] EXAMPLE EMBODIMENT 12. A chiplet comprising:

[0080] a die having a first communication channel; and

[0081] a first interface associated with the first communication channel, wherein the interface comprises:

[0082] circuitry configured to support an advanced interface bus 2.0 (AIB 2.0) protocol connection; and

[0083] a plurality of bumps having a density less dense than specified for the AIB 2.0 protocol, wherein the first interface is configured to communicate using the AIB 2.0 protocol.

[0084] EXAMPLE EMBODIMENT 13. The chiplet of example embodiment 12, comprising a redistribution layer that facilitates connection between the plurality of bumps and the die.

[0085] EXAMPLE EMBODIMENT 14. The chiplet of example embodiment 13, wherein the chiplet is communicatively coupled to a second chiplet via the interface.

[0086] EXAMPLE EMBODIMENT 15. The chiplet of example embodiment 12, wherein the interface is configured to have a throughput that is a fraction of a throughput corresponding to the AIB 2.0 protocol.

[0087] EXAMPLE EMBODIMENT 16. The chiplet of example embodiment 12, wherein the bumps comprise C4 bumps.

[0088] EXAMPLE EMBODIMENT 17. The chiplet of example embodiment 12, wherein the die comprises a second channel comprising a second interface comprising an additional plurality of bumps having a density specified for the AIB 2.0 protocol, and wherein the second interface is configured to communicate using the AIB 2.0 protocol.

[0089] EXAMPLE EMBODIMENT 18. A chip comprising:

[0090] a top metal layer; and

[0091] a redistribution layer, wherein the redistribution layer is configured to selectively use a first interface layer with a first communication architecture or a second interface layer with a second communication architecture as the top metal layer, wherein the first interface layer comprises a fraction of a number of bumps of the second interface layer, wherein the first communication architecture and the second communication architecture are configured to communicate using a same communication protocol.

[0092] EXAMPLE EMBODIMENT 19. The chip of example embodiment 18, wherein the same communication protocol comprises an advanced interface bus (AIB).

[0093] EXAMPLE EMBODIMENT 20. The chip of example embodiment 18, wherein the first interface layer comprises micro-bumps and the second interface layer comprises C4 bumps.

What is claimed is:

**1**. A semiconductor device comprising:

a first die having a first communication channel; and

a first interface associated with the first communication channel, wherein the first interface comprises:

circuitry configured to couple to a second interface having a higher bandwidth compared to that of the first interface; and

a first plurality of bumps having a density less dense than specified for the second interface, wherein the first die is configured to communicate using a first protocol and a second protocol via the first interface, wherein the second protocol is associated with the second interface.

**2**. The semiconductor device of claim **1**, wherein the second interface having the higher bandwidth is associated with an advanced interface bus 2.0 (AIB 2.0) protocol connection.

**3**. The semiconductor device of claim **1**, wherein the second protocol corresponds to an AIB 2.0 protocol connection.

**4**. The semiconductor device of claim **1**, wherein the first protocol corresponds to an AIB-O protocol connection.

**5**. The semiconductor device of claim **1**, wherein the first die is configured to couple to and communicate with a second die having the second interface based on coupling the first interface to the second interface.

**6**. The semiconductor device of claim **1**, comprising a redistribution layer that facilitates connection between the first plurality of bumps and the first die.

**7**. The semiconductor device of claim **1**, wherein the first interface is configured to have a throughput that is a fraction of a throughput of the second interface.

**8**. The semiconductor device of claim **1**, wherein the first die has a first communication channel comprising the first interface.

**9**. The semiconductor device of claim **8**, wherein the first die has a second communication channel comprising a third interface having a higher bandwidth compared to that of the first interface and a second plurality of bumps having a density corresponding to that specified for the second interface, wherein the first die is configured to communicate using the second protocol via the second interface.

**10**. The semiconductor device of claim **1**, wherein the first plurality of bumps comprises C4 bumps.

**11**. A chiplet comprising:

a first die having a first communication channel; and

a first interface associated with the first communication channel, wherein the first interface comprises:

circuitry configured to support an advanced interface bus 2.0 (AIB 2.0) protocol connection; and

a plurality of bumps having a density less dense than specified for the AIB 2.0 protocol, wherein the first interface is configured to communicate using the AIB 2.0 protocol.

**12**. The chiplet of claim **11**, comprising a redistribution layer that facilitates connection between the plurality of bumps and the first die.

**13**. The chiplet of claim **12**, wherein the chiplet is communicatively coupled to a second chiplet via the first interface.

**14**. The chiplet of claim **13**, wherein the first interface is configured to have a throughput that is a fraction of a throughput corresponding to the AIB 2.0 protocol.

**15**. The chiplet of claim **13**, wherein the plurality of bumps comprise C4 bumps.

**16**. The chiplet of claim **13**, wherein the first die has a second communication channel comprising a second interface comprising an additional plurality of bumps having a density specified for the AIB 2.0 protocol, and wherein the second interface is configured to communicate using the AIB 2.0 protocol.

**17**. The chiplet of claim **11**, wherein the first die is configured to couple to and communicate with a third interface of a second die comprising an additional plurality of bumps having a density specified for the AIB 2.0 protocol.

**18**. A chip comprising:

a first interface layer;

a redistribution layer coupled to the first interface layer, wherein the redistribution layer is coupled to the first interface layer; and

a first die coupled to the redistribution layer, wherein the die is configured to communicate with a second die using the first interface layer with a first communication architecture and using a second interface layer with a second communication architecture as a top metal layer, wherein the first interface layer comprises a fraction of a number of bumps of the second interface layer, wherein the first communication architecture and the second communication architecture are configured to communicate using a same communication protocol.

**19**. The chip of claim **18**, wherein the same communication protocol comprises an advanced interface bus (AIB).

**20**. The chip of claim **18**, wherein the first interface layer comprises micro-bumps and the second interface layer comprises C4 bumps.

* * * * *